

Aura & Petals System Design Document

1. Project Overview & Architecture

Aura & Petals is a modern, production-level, multi-tenant capable E-commerce and SaaS platform. It automates product distribution across multiple social media platforms while serving an aesthetic, e-commerce storefront for end-users. The architecture consists of a storefront, an admin panel for small and medium-sized enterprises (SMEs), and a super-admin panel for SaaS management.

2. Tech Stack & Infrastructure

Layer	Technology
Frontend	Next.js (App Router), Tailwind CSS, Framer Motion
Backend / BaaS	Supabase (Database, Auth, Storage)
Hosting	Vercel
Cloud Image/Video Uploads	Cloudinary
Automation & Webhooks	Make.com

3. Frontend Storefront Architecture

- **Home Page:** Hero section with banner, featured items, and category links.
- **Shop/Products Page:** Grid display with search bar, sorting, and multi-parameter filtering.
- **Product Detail Page:** Multiple images, pricing, and specs.
- **Customer Authentication & Profile:** Login/Register, profile settings, and order history tracking.
- **Contact Page:** Contact form integrated with the Admin message board.

4. Admin Dashboard (SME Management)

Admin Control Panel Features:

- **Branding Settings:** Upload logo and change primary/secondary hex colors.
- **Product Management:** Create, Read, Update, and Delete operations.
- **Image Handling:** Cloudinary integration with automatic cleanup on product deletion.
- **Messages:** View and manage incoming messages from the contact page.

5. Super-Admin (SaaS Owner) Dashboard

- **Subscription Management:** Activate, modify, or disable client subscriptions.
- **Pricing Plans:** Manage plan tiers for different business sizes.
- **Payment Processing:** Manual recording and management of bKash and Nagad transactions.

6. Automation Workflow & Rules

Image and Video Distribution Rules

Distribution Flow:

1. Trigger: Product Create / Update in Supabase
2. Condition 1: If Video (Shorts/Reels only) -> Send to Facebook, Instagram, YouTube, TikTok, LinkedIn.
3. Condition 2: If Image -> Send to Facebook, Instagram, LinkedIn.
4. Webhook: Next.js or Supabase triggers Make.com payload.

- **Image Usage:** Appears everywhere except YouTube and TikTok. Multiple images can be uploaded and managed.
- **Video Usage:** Short-form content only (Reels/Shorts). Videos are not displayed on the e-commerce website.
- **Cloudinary Sync:** Cloudinary handles image updates and deletes to keep the storage clean.